Database Systems

(Course Code: **CSC270**)

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Theory Assignment# 2

Problem 1: Bioinformatics Database

The bioinformatics system manages patients along with their tissue sample tag libraries. Each tag may be linked to a gene, and genes can be referenced in multiple research articles, introducing several many-to-many relationships. Additionally, the relationship between patients and tags includes an extra attribute (count), which requires modeling it as an associative entity rather than a simple join table.

1.1 ER Diagram

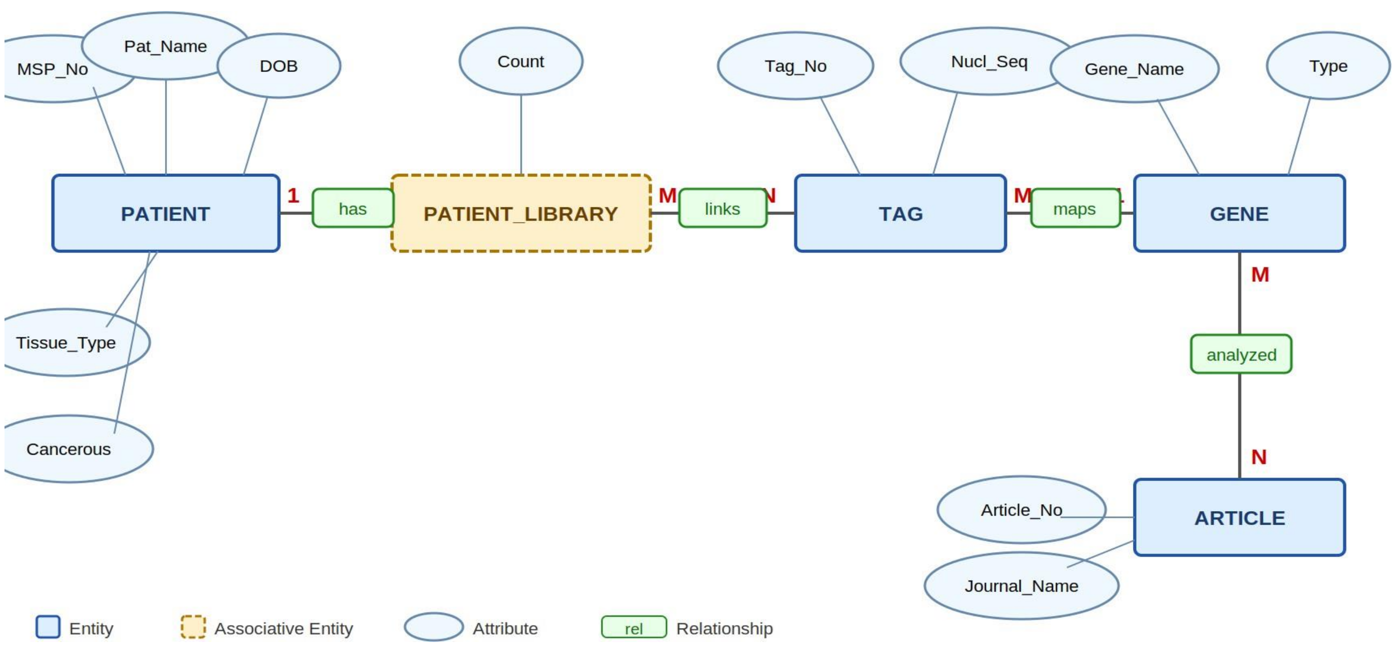

Figure 1: ER Diagram – Bioinformatics Database

1.2 Relational Schema

The following tables are derived from the ER diagram, where primary keys (PK) uniquely identify each record and foreign keys (FK) establish relationships between the tables.

PATIENT

Attribute	Type	Constraint
MSP_No	VARCHAR(20)	**PK – Unique patient identifier**
Patient_Name	VARCHAR(100)	NOT NULL

Date_of_Birth	DATE	NOT NULL
Tissue_Type	VARCHAR(50)	NOT NULL
Is_Cancerous	CHAR(1)	CHECK ('Y','N') – indicates cancer status

TAG

Attribute	Type	Constraint
Tag_No	VARCHAR(20)	PK – Unique tag identifier
Nucleotide_Sequence	TEXT	NOT NULL UNIQUE
Gene_Name	VARCHAR(100)	FK → GENE(Gene_Name) – nullable (tag may not be mapped)

GENE

Attribute	Type	Constraint
Gene_Name	VARCHAR(100)	PK – Unique gene name
Gene_Type	VARCHAR(50)	NOT NULL

ARTICLE

Attribute	Type	Constraint
Article_No	VARCHAR(20)	PK – Unique article identifier
Journal_Name	VARCHAR(150)	NOT NULL

PATIENT_LIBRARY (Associative – stores tag count per patient)

Attribute	Type	Constraint
MSP_No	VARCHAR(20)	PK, FK → PATIENT(MSP_No)
Tag_No	VARCHAR(20)	PK, FK → TAG(Tag_No)
Count	INT	NOT NULL – occurrences of tag in library

GENE_ARTICLE (M:N between GENE and ARTICLE)

Attribute	Type	Constraint
Gene_Name	VARCHAR(100)	PK, FK → GENE(Gene_Name)

Article_No	VARCHAR(20)	PK, FK → ARTICLE(Article_No)

Problem 2: Blue Wave Water Sports Center Database

Blue Wave operates two distinct services—Jet Ski rentals and Surfboard sales. While both equipment types share a common attribute (Model Name), they differ in pricing structure (hourly rental vs. fixed sale price). Similarly, tourists share general attributes like contact information but differ in role-specific details. These variations are best modeled using ISA hierarchies to maintain a clean and structured design.

2.1 ER Diagram

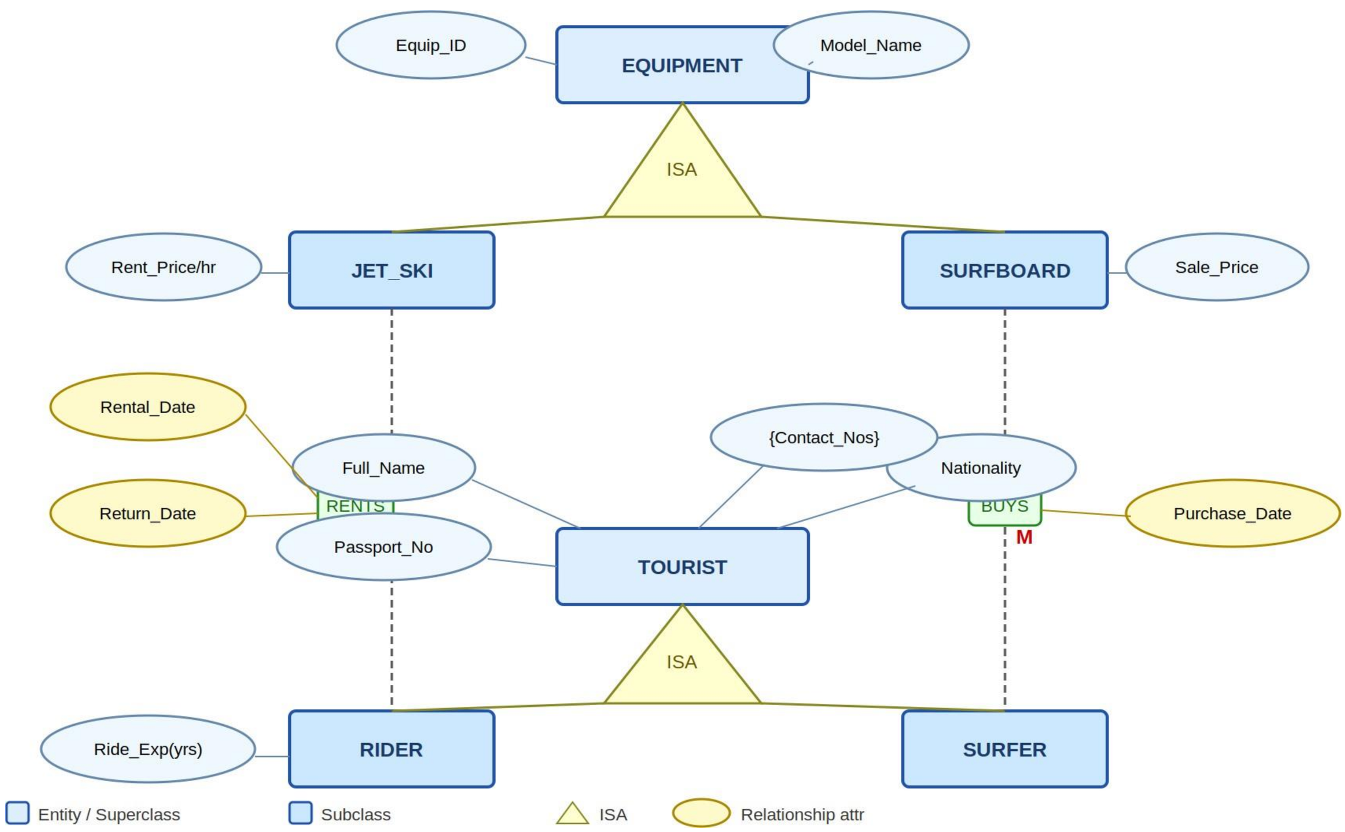

Figure 2: ER Diagram – Blue Wave Water Sports Center

2.2 Relational Schema

EQUIPMENT (Superclass)

Attribute	Type	Constraint
Equipment_ID	VARCHAR(20)	PK
Model_Name	VARCHAR(100)	NOT NULL
Equip_Type	CHAR(1)	CHECK('J','S') – J=JetSki, S=Surfboard

JET_SKI (Subclass of EQUIPMENT)

Attribute	Type	Constraint
Equipment_ID	VARCHAR(20)	PK, FK → EQUIPMENT(Equipment_ID)
Rental_Price_Per_Hour	DECIMAL(8,2)	NOT NULL

SURFBOARD (Subclass of EQUIPMENT)

Attribute	Type	Constraint
Equipment_ID	VARCHAR(20)	PK, FK → EQUIPMENT(Equipment_ID)
Sale_Price	DECIMAL(8,2)	NOT NULL

TOURIST (Superclass)

Attribute	Type	Constraint
Passport_No	VARCHAR(30)	PK
Full_Name	VARCHAR(100)	NOT NULL
Nationality	VARCHAR(60)	NOT NULL
Tourist_Type	CHAR(1)	CHECK('R','S') – R=Rider, S=Surfer

TOURIST_CONTACT (Multi valued attribute)

Attribute	Type	Constraint
Passport_No	VARCHAR(30)	PK, FK → TOURIST(Passport_No)
Contact_Number	VARCHAR(20)	PK

RIDER (Subclass of TOURIST)

Attribute	Type	Constraint
Passport_No	VARCHAR(30)	PK, FK → TOURIST(Passport_No)
Riding_Experience_Years	INT	NOT NULL

SURFER (Subclass of TOURIST)

Attribute	Type	Constraint

Passport_No	VARCHAR(30)	PK, FK → TOURIST(Passport_No)

RENTAL (M:N between RIDER and JET_SKI)

Attribute	Type	Constraint
Passport_No	VARCHAR(30)	PK, FK → RIDER(Passport_No)
Equipment_ID	VARCHAR(20)	PK, FK → JET_SKI(Equipment_ID)
Rental_Date	DATE	PK – part of composite key
Return_Date	DATE	NOT NULL

PURCHASE (M:1 between SURFER and SURFBOARD)

Attribute	Type	Constraint
Equipment_ID	VARCHAR(20)	PK, FK → SURFBOARD(Equipment_ID)
Passport_No	VARCHAR(30)	FK → SURFER(Passport_No)
Purchase_Date	DATE	NOT NULL

Problem 3: Football Match Database

This is the most entity-intensive scenario among the five. Football clubs serve as the central hub, connecting to games, personnel, uniforms, jargon, reference materials, deliverables, and value systems. A key design decision is modeling **Game_Player** as an associative entity, since a player's position depends on a specific game context rather than being a fixed attribute of the player.

3.1 ER Diagram

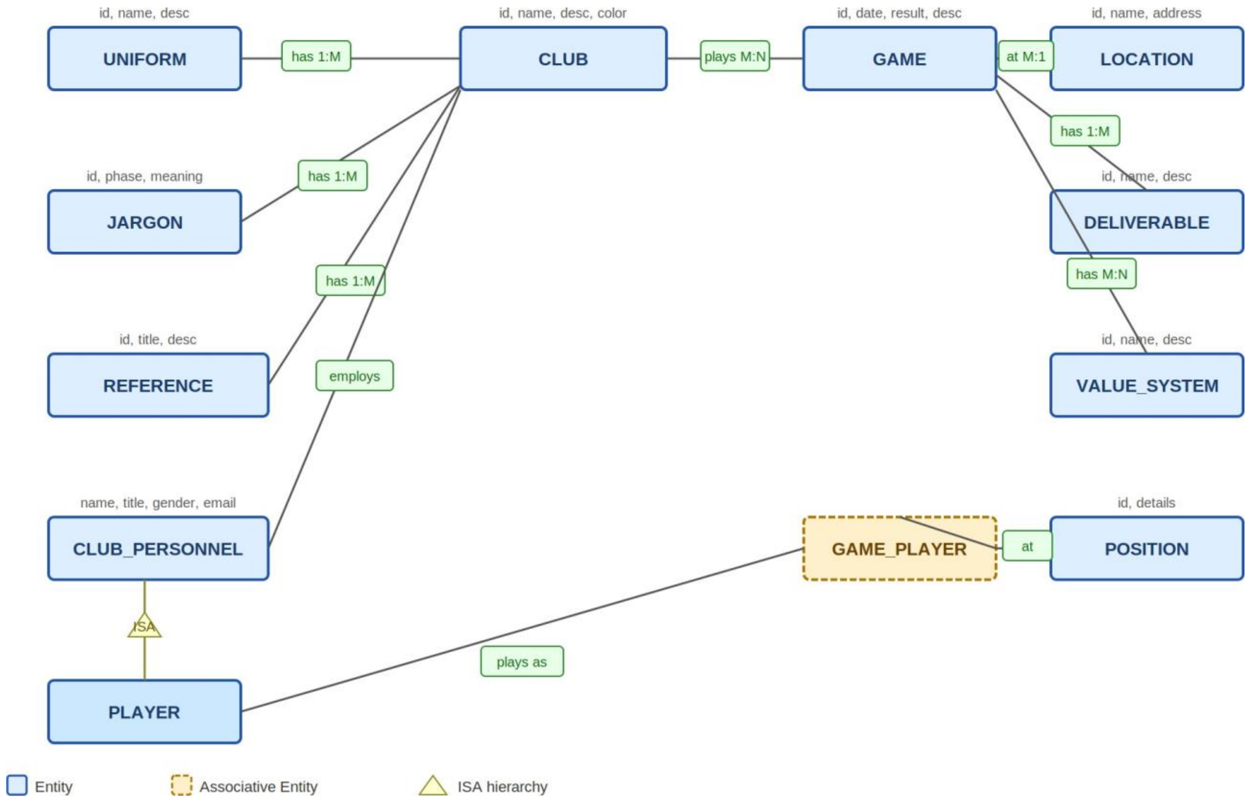

Figure 3: ER Diagram – Football Match Database

3.2 Relational Schema

CLUB

Attribute	Type	Constraint
Club_ID	VARCHAR(20)	PK
Club_Name	VARCHAR(100)	NOT NULL UNIQUE
Description	TEXT	
Color	VARCHAR(50)	

UNIFORM

Attribute	Type	Constraint
Uniform_ID	VARCHAR(20)	**PK**
Uniform_Name	VARCHAR(100)	NOT NULL
Description	TEXT	
Club_ID	VARCHAR(20)	**FK → CLUB(Club_ID)**

JARGON

Attribute	Type	Constraint
Jargon_ID	VARCHAR(20)	PK
Phase	VARCHAR(200)	NOT NULL
Meaning	TEXT	NOT NULL
Club_ID	VARCHAR(20)	FK → CLUB(Club_ID)

REFERENCE_MATERIAL

Attribute	Type	Constraint
Ref_ID	VARCHAR(20)	PK
Title	VARCHAR(200)	NOT NULL
Description	TEXT	
Club_ID	VARCHAR(20)	FK → CLUB(Club_ID)

CLUB_PERSONNEL

Attribute	Type	Constraint
Personnel_ID	VARCHAR(20)	PK
Name	VARCHAR(100)	NOT NULL
Title	VARCHAR(60)	
Gender	VARCHAR(10)	
Address	VARCHAR(200)	
Email	VARCHAR(100)	
Contact	VARCHAR(30)	
Club_ID	VARCHAR(20)	FK → CLUB(Club_ID)

PLAYER (Subclass of CLUB_PERSONNEL)

Attribute	Type	Constraint
Personnel_ID	VARCHAR(20)	PK, FK → CLUB_PERSONNEL(Personnel_ID)

GAME

Attribute	Type	Constraint
Game_ID	VARCHAR(20)	PK
Game_Date	DATE	NOT NULL
Results	VARCHAR(50)	
Description	TEXT	
Location_ID	VARCHAR(20)	FK → LOCATION(Location_ID)

CLUB_GAME (M:N – two clubs per game)

Attribute	Type	Constraint
Game_ID	VARCHAR(20)	PK, FK → GAME(Game_ID)
Club_ID	VARCHAR(20)	PK, FK → CLUB(Club_ID)
Role	VARCHAR(20)	CHECK('Home','Away')

LOCATION

Attribute	Type	Constraint
Location_ID	VARCHAR(20)	PK
Location_Name	VARCHAR(100)	NOT NULL
Address	VARCHAR(200)	

DELIVERABLE

Attribute	Type	Constraint
Deliverable_ID	VARCHAR(20)	PK
Name	VARCHAR(100)	NOT NULL
Description	TEXT	
Game_ID	VARCHAR(20)	FK → GAME(Game_ID)

VALUE_SYSTEM

Attribute	Type		Constraint	

VS_ID	VARCHAR(20)	PK
Name	VARCHAR(100)	NOT NULL
Description	TEXT	

GAME_VALUE_SYSTEM

Attribute	**Type**	**Constraint**
Game_ID	VARCHAR(20)	PK, FK → GAME(Game_ID)
VS_ID	VARCHAR(20)	PK, FK → VALUE_SYSTEM(VS_ID)

POSITION

Attribute	**Type**	**Constraint**
Position_ID	VARCHAR(20)	PK
Position_Details	TEXT	

GAME_PLAYER (Associative – Player in Game at Position)

Attribute	**Type**	**Constraint**
Game_ID	VARCHAR(20)	PK, FK → GAME(Game_ID)
Personnel_ID	VARCHAR(20)	PK, FK → PLAYER(Personnel_ID)
Position_ID	VARCHAR(20)	FK → POSITION(Position_ID)

Problem 4: Chughtai Group of Pharmacies Database

The most challenging aspect of this design is the DRUG table. Since trade names are only unique per company and not globally, the table requires a composite primary key consisting of (Trade_Name + Company_Name). This composite key then carries over into both the PRESCRIPTION and PHARMACY_DRUG tables. All other elements adhere to standard normalization principles.

4.1 ER Diagram

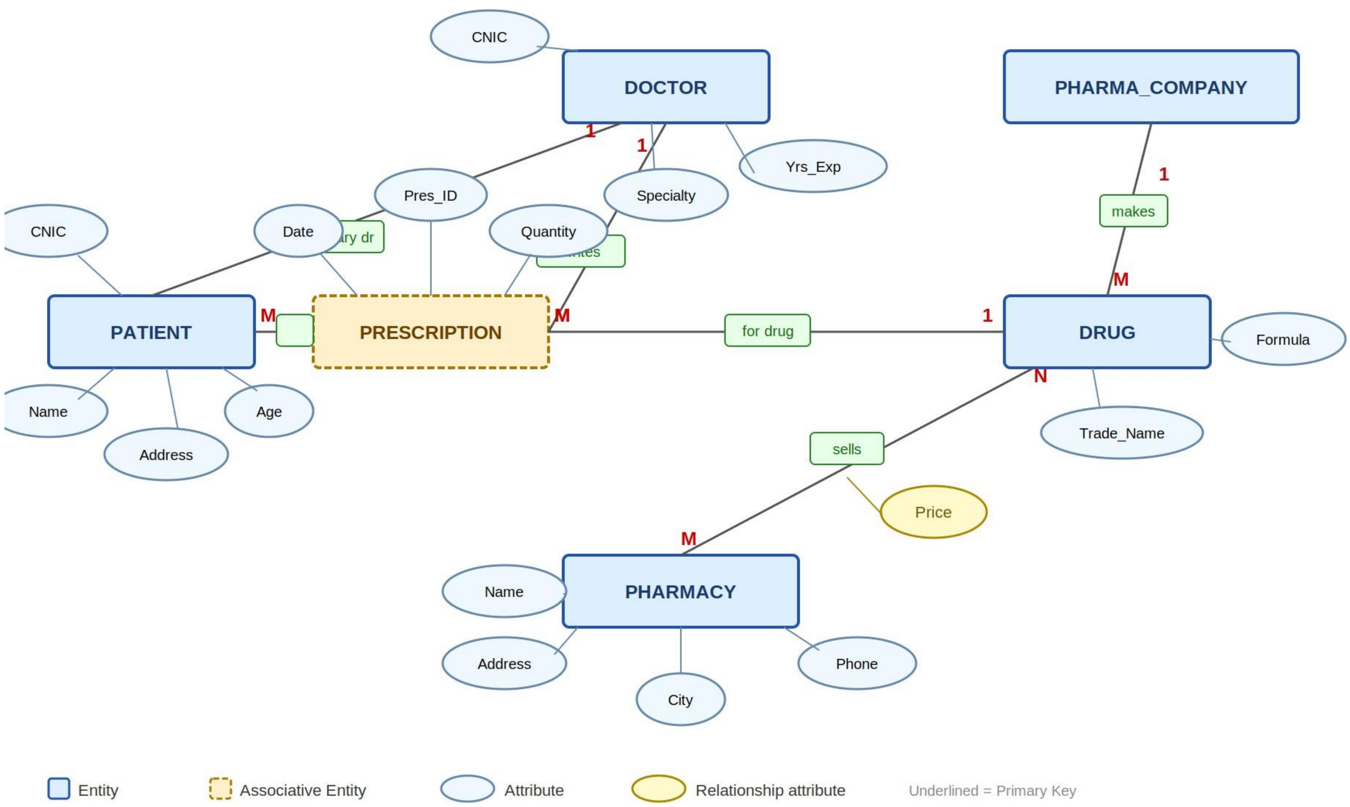

Figure 4: ER Diagram – Chughtai Pharmacies Database

4.2 Relational Schema

PATIENT

Attribute	Type	Constraint
CNIC	CHAR(13)	PK
Name	VARCHAR(100)	NOT NULL
Address	VARCHAR(200)	
Age	INT	CHECK(Age > 0)

Doctor_CNIC	CHAR(13)	FK → DOCTOR(CNIC) – primary doctor (M:1)

DOCTOR

Attribute	Type	Constraint
CNIC	CHAR(13)	PK
Name	VARCHAR(100)	NOT NULL
Specialty	VARCHAR(100)	NOT NULL
Years_of_Experience	INT	CHECK >= 0

PHARMA_COMPANY

Attribute	Type	Constraint
Company_Name	VARCHAR(150)	PK
Phone_Number	VARCHAR(20)	NOT NULL

DRUG (Trade_Name unique per company only → composite PK)

Attribute	Type	Constraint
Trade_Name	VARCHAR(100)	PK (part of composite)
Company_Name	VARCHAR(150)	PK (part of composite), FK → PHARMA_COMPANY(Company_Name)
Formula	VARCHAR(200)	NOT NULL

PHARMACY

Attribute	Type	Constraint
Pharmacy_Name	VARCHAR(150)	PK
Address	VARCHAR(200)	NOT NULL
City	VARCHAR(60)	NOT NULL
Phone_Number	VARCHAR(20)	NOT NULL

PHARMACY_DRUG (M:N – drug sold at pharmacy with price)

Attribute	Type	Constraint
Pharmacy_Name	VARCHAR(150)	PK, FK → PHARMACY(Pharmacy_Name)
Trade_Name	VARCHAR(100)	PK (part of FK → DRUG)

Company_Name	VARCHAR(150)	PK (part of FK → DRUG)
Price	DECIMAL(8,2)	NOT NULL – may vary per pharmacy

PRESCRIPTION (Ternary: Doctor prescribes Drug for Patient)

Attribute	Type	Constraint
Prescription_ID	VARCHAR(20)	PK
Prescription_Date	DATE	NOT NULL
Quantity	INT	NOT NULL
Doctor_CNIC	CHAR(13)	FK → DOCTOR(CNIC)
Patient_CNIC	CHAR(13)	FK → PATIENT(CNIC)
Trade_Name	VARCHAR(100)	FK (part) → DRUG
Company_Name	VARCHAR(150)	FK (part) → DRUG

Problem 5: University Database

The university schema enforces clear role separation. Assistant Professors teach up to five courses, Associate Professors oversee seminars, and Professors manage departments and handle funded projects. These three roles are represented as disjoint subclasses of a common FACULTY superclass.

5.1 ER Diagram

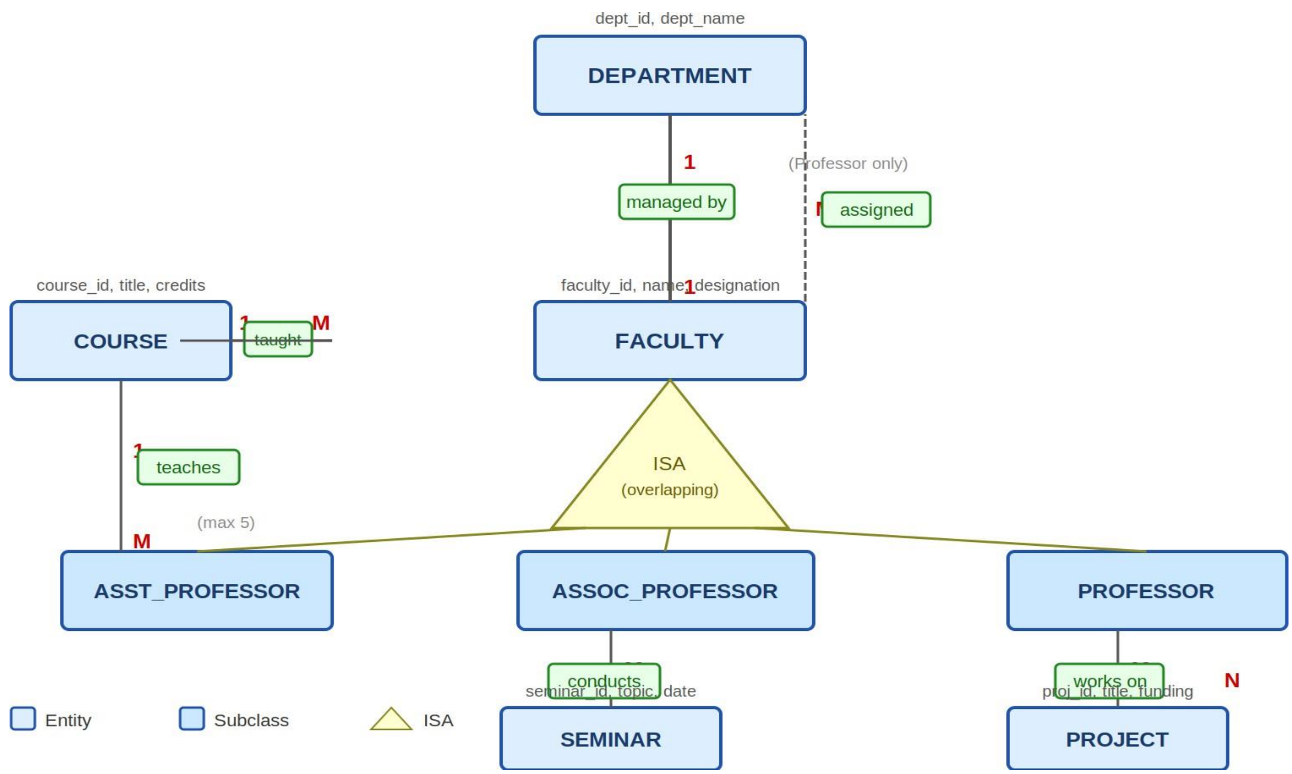

Figure 5: ER Diagram – University Database

5.2 Key Business Rules

- A Professor manages each department one professor, one department (1:1).
- Every faculty member belongs to exactly one department.
- Only Assistant Professors teach courses. One course has at most one instructor; one instructor handles at most five courses.
- Only Associate Professors conduct seminars.
- Only Professors work on funded projects (M:N — a professor can run multiple projects).

5.3 Relational Schema

DEPARTMENT

Attribute	Type	Constraint
Dept_ID	VARCHAR(20)	PK
Dept_Name	VARCHAR(100)	NOT NULL UNIQUE
Manager_Faculty_ID	VARCHAR(20)	FK → PROFESSOR(Faculty_ID) – 1:1, NOT NULL

FACULTY (Superclass)

Attribute	Type	Constraint
Faculty_ID	VARCHAR(20)	PK
Name	VARCHAR(100)	NOT NULL
Designation	VARCHAR(20)	CHECK IN ('Asst_Prof','Assoc_Prof','Professor')
Dept_ID	VARCHAR(20)	FK → DEPARTMENT(Dept_ID) – each faculty in one department

ASST_PROFESSOR (Subclass of FACULTY)

Attribute	Type	Constraint
Faculty_ID	VARCHAR(20)	PK, FK → FACULTY(Faculty_ID)

ASSOC_PROFESSOR (Subclass of FACULTY)

Attribute	Type	Constraint
Faculty_ID	VARCHAR(20)	PK, FK → FACULTY(Faculty_ID)

PROFESSOR (Subclass of FACULTY)

Attribute	Type	Constraint
Faculty_ID	VARCHAR(20)	PK, FK → FACULTY(Faculty_ID)

COURSE

Attribute	Type	Constraint
Course_ID	VARCHAR(20)	PK

Title	VARCHAR(150)	NOT NULL
Credits	INT	DEFAULT 3
Faculty_ID	VARCHAR(20)	FK → ASST_PROFESSOR(Faculty_ID) – nullable (unassigned)

SQL can't directly enforce that an Assistant Professor teaches no more than 5 courses a foreign key can't count.

SEMINAR

Attribute	**Type**	**Constraint**
Seminar_ID	VARCHAR(20)	PK
Topic	VARCHAR(200)	NOT NULL
Seminar_Date	DATE	
Faculty_ID	VARCHAR(20)	FK → ASSOC_PROFESSOR(Faculty_ID) – only Assoc. Profs

PROJECT

Attribute	**Type**	**Constraint**
Project_ID	VARCHAR(20)	PK
Title	VARCHAR(200)	NOT NULL
Funding_Amount	DECIMAL(12,2)	
Funding_Source	VARCHAR(150)	

PROFESSOR_PROJECT (M:N – Professors work on Projects)

Attribute	**Type**	**Constraint**
Faculty_ID	VARCHAR(20)	PK, FK → PROFESSOR(Faculty_ID)
Project_ID	VARCHAR(20)	PK, FK → PROJECT(Project_ID)